

Not just design

Marvell's Navin Bishnoi on next chapter in Indian semicon

Jayesh Shinde | editor@digit.in

For three decades, the story of India's contribution in semiconductors has almost entirely been viewed from a design lens. That framing is now starting to look outdated, and few people are better positioned to explain the shift underway than Navin Bishnoi, VP and India Country Manager at Marvell Technology and currently chairs the India Electronics and Semiconductor Association (IESA).

A fabless designer of AI and data-infrastructure silicon, Marvell's one of the key custom-ASIC partners to the world's hyperscalers, and it just completed 20 years in the country, mirroring India's own evolution from offshore design centre hub to an aspiring product nation. In an exclusive chat, Bishnoi laid out why he believes India's semiconductor journey is moving "from execution to product ownership," how the AI boom is forcing a fundamental rewiring of the datacentre, and why the rush of every big-tech firm to build its own custom chip is an incredible opportunity for semiconductor companies and India. Edited excerpts follow:

Q Where does India fit into the global semiconductor value chain beyond just being a design hub?

A Navin Bishnoi: We have been in

the ecosystem for four decades, and a large part of that focus has been on the design ecosystem. What we are seeing over the last five years is a balance on accelerating the manufacturing footprint, primarily driven

by resilience and overall value chain improvement. We are seeing an acceleration toward end-to-end product creation – right from idea and architecture to design, validation, systems, and software. We need to create more differentiated or foundational IP, have more system-level innovation, and leverage both the local and global ecosystems to drive designed manufacturing.



Navin Bishnoi
VP & India Country
Head, Marvell
Technology

Q What are some of the things India's semiconductor ecosystem is getting right, and what are the areas for improvement?

A Navin Bishnoi: There are three or four things that have started falling into place. The strongest is the stronger policy intent. We saw the impact ISM 1.0 created. The ecosystem is bubbling – not just with announcements, but with execution as well. Second is the acknowledgment that this will require bigger collaboration across different bodies, which includes not just government and industry bodies, but academia and research labs. And third, while the

talent has been very focused on the design side, we need to scale into full ecosystem enablement.

What we need to improve is how we solve the next problem. There is still scope to ask: how do startups get access to patient, deep-tech capital? And how do they get a stronger anchor demand – a path to market – so they can sell and then scale? A good thing is that a lot of this feedback is going into the 2.0 version of ISM. We probably need to not just double, but triple and quadruple the number of startup outcomes.

Q From a semiconductor supply chain perspective, what key inroads can India make in the next 5-10 years?

A Navin Bishnoi: The global environment has reinforced the importance of resilience, compliance, and a diversified innovation ecosystem. For many global semiconductor companies, including ours, that means building very strong R&D and execution capabilities across multiple regions. The large talent pool here is very deep-tech, with skills across product development, from practitioners to architects. And hence, what was an execution center has been converted into a major R&D base and, in many cases, a product creation hub.

Having deep researchers who can look not just at what's needed today, but tomorrow and the day after – I call it an N plus one and N plus two vision of IP and tech – the goal is to move from execution into product ownership. So 5, 10, or 15 years down the line, I see critical product hubs of large companies, along with very niche, competitive startups and products from here, trying to solve what we need in digital infrastructure – not just for India or the Global South, but from a global ownership perspective. **d**

Read more on Digit.in:
tinyurl.com/yc7j3v84